

API Documentation

Healthcare Patient Management System, version 0.1.0. Interactive Swagger UI:

<http://127.0.0.1:8000/docs> ; machine-readable contract: <http://127.0.0.1:8000/openapi.json> .

This document is generated by `scripts/make_api_docs.py` from the running application's OpenAPI document and from real calls against the development database (101,766 encounters). Admission and discharge dates in this data are **simulated**; every readmission rate uses *eligible* encounters (expired and hospice discharges excluded) as the denominator. Educational use only.

1. Conventions

Topic	Rule
Authentication	POST <code>/auth/login</code> (form fields <code>username</code> , <code>password</code>) returns a bearer token (HS256 JWT, valid 60 minutes). Send <code>Authorization: Bearer <token></code> . In the samples below <code>\$TOKEN</code> is that token.
Roles	<code>administrator</code> , <code>clinical_ops</code> , <code>analyst</code> . Roles are enforced on the server for every request; the table in section 2 is the single source (<code>app/permissions.py</code>).
Error body	<code>{"error": {"code": "...", "message": "...", "details": ...}}</code> for every error. Common codes: 401 <code>unauthorized</code> , 403 <code>forbidden</code> , 404 <code>not_found</code> , 409 <code>conflict</code> , 422 <code>validation_error</code> , 413 <code>too_large</code> . No stack trace or SQL is ever returned.
Request id	Every response carries <code>X-Request-ID</code> ; the same id is on each log line for that request.
Lists	<code>{"items": [...], "total": n, "page": n, "page_size": n, "pages": n, "meta": {...}}</code> . <code>page</code> starts at 1, <code>page_size</code> defaults to 25 and is capped at 100. Sorting uses a whitelist of columns plus a unique tie-breaker, so pages never repeat or skip rows.
Deletes	<code>DELETE</code> is a soft delete (<code>is_deleted</code>): the record disappears from lists, analytics and exports, and the change is audited.
Analytics	Responses are <code>{"data": ..., "meta": {...}}</code> . <code>meta</code> states the denominator and <code>dates_simulated: true</code> . Groups with fewer than 11 encounters are flagged <code>small_n</code> .
Audit	Every create, update, delete and export writes an <code>audit_logs</code> row in the same database transaction as the change.

2. Who can call what (permission matrix)

Capability	Roles allowed
<code>dashboards</code>	<code>administrator</code> , <code>analyst</code> , <code>clinical_ops</code>
<code>export_aggregated</code>	<code>administrator</code> , <code>analyst</code> , <code>clinical_ops</code>
<code>list_encounters</code>	<code>administrator</code> , <code>analyst</code> , <code>clinical_ops</code>
<code>export_encounters</code>	<code>administrator</code> , <code>clinical_ops</code>
<code>view_patients</code>	<code>administrator</code> , <code>clinical_ops</code>
<code>write_records</code>	<code>administrator</code> , <code>clinical_ops</code>
<code>delete_records</code>	<code>administrator</code>
<code>view_pipeline</code>	<code>administrator</code> , <code>clinical_ops</code>

Capability	Roles allowed
view_audit	administrator
manage_users	administrator

Analysts calling `GET /encounters` receive a minimised response without `patient_nbr`, race or gender.

3. Endpoints

Method	Path	Roles	Summary
POST	/auth/login	public (no token)	Log in and receive a bearer token
GET	/auth/me	any signed-in user	The authenticated user
GET	/health	public (no token)	Service and database status
POST	/patients	administrator, clinical_ops	Create a patient
GET	/patients	administrator, clinical_ops	List and search patients
GET	/patients/{patient_nbr}	administrator, clinical_ops	Get one patient
PUT	/patients/{patient_nbr}	administrator, clinical_ops	Update a patient (partial)
DELETE	/patients/{patient_nbr}	administrator	Soft-delete a patient (administrator only)
GET	/patients/{patient_nbr}/encounters	administrator, clinical_ops	Active encounters of one patient
POST	/encounters	administrator, clinical_ops	Create an encounter
GET	/encounters	administrator, analyst, clinical_ops	List and search encounters
GET	/encounters/{encounter_id}	administrator, analyst, clinical_ops	Get one encounter with diagnoses and medications
PUT	/encounters/{encounter_id}	administrator, clinical_ops	Update an encounter (partial)
DELETE	/encounters/{encounter_id}	administrator	Soft-delete an encounter (administrator only)
GET	/analytics/summary	administrator, analyst, clinical_ops	Headline KPIs
GET	/analytics/admissions-trend	administrator, analyst, clinical_ops	Encounters and readmission rate per month or year
GET	/analytics/length-of-stay	administrator, analyst, clinical_ops	Length of stay by group, with histogram
GET	/analytics/readmissions	administrator, analyst, clinical_ops	30-day readmission rate by group
GET	/analytics/medications	administrator, analyst, clinical_ops	Top drugs, insulin status and A1C vs readmission
GET	/analytics/utilization	administrator, analyst, clinical_ops	Prior inpatient visits vs readmission

Method	Path	Roles	Summary
GET	/reports/export	aggregated reports: all roles; report=encounters : administrator, clinical_ops	Download a report as CSV or Excel
GET	/reports/history	administrator, clinical_ops	Generated reports (administrator, clinical_ops)
GET	/admin/audit-logs	administrator	Audit trail (administrator)
GET	/admin/pipeline-runs	administrator, clinical_ops	ETL run history
GET	/admin/dq-issues	administrator, clinical_ops	Row-level data-quality findings
GET	/admin/users	administrator	List users (never includes password hashes)
POST	/admin/users	administrator	Create a user (administrator)
GET	/reference/admission-types	any signed-in user	Admission Types
GET	/reference/admission-sources	any signed-in user	Admission Sources
GET	/reference/discharge-dispositions	any signed-in user	Discharge Dispositions
GET	/reference/specialties	any signed-in user	Specialties
GET	/reference/age-groups	any signed-in user	Age groups in clinical order (id = age_order)

3.1 POST /auth/login

Log in and receive a bearer token.

Roles allowed: public (no token)

Request body (application/x-www-form-urlencoded)

Field	Type	Required
username	string	yes
password	string	yes

Response format: success 200, body TokenOut . Declared status codes: 200, 422 .

Sample (real call)

Request ((no token)):

```
curl -X POST -d "username=docs_administrator_73ac07" -d "password=<password>" "http://127.0.0.1:8000/auth/login"
```

Response 200 :

```
{
  "access_token": "eyJhbGciOiJIUzI1Ni...cAs06s",
  "token_type": "bearer",
  "expires_in": 3600
}
```

3.2 GET /auth/me

The authenticated user.

Roles allowed: any signed-in user

Response format: success 200, body UserOut . Declared status codes: 200 .

Sample (real call)

Request (administrator):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/auth/me"
```

Response 200:

```
{
  "user_id": 8,
  "username": "docs_administrator_73ac07",
  "role": "administrator"
}
```

3.3 GET /health

Service and database status.

Roles allowed: public (no token)

Response format: success 200, no body. Declared status codes: 200 .

Sample (real call)

Request ((no token)):

```
curl -X GET "http://127.0.0.1:8000/health"
```

Response 200:

```
{
  "status": "ok",
  "database": "up",
  "latest_pipeline_run": {
    "run_id": 13,
    "source_file": "encounters_batch_003.csv",
    "status": "SUCCESS",
    "started_at": "2026-10-09T20:54:06",
    "finished_at": "2026-10-09T20:54:06",
    "rows_read": 5088,
    "rows_loaded": 23,
    "rows_rejected": 0,
    "rows_skipped_existing": 5065
  }
}
```

3.4 POST /patients

Create a patient.

Roles allowed: administrator, clinical_ops

Request body (application/json)

Field	Type	Required
patient_nbr	integer (maximum 9.223372036854776e+18, exclusiveMinimum 0.0)	yes
race	string (maxLength 30)	no
gender	one of: Male, Female, Unknown	no

Response format: success 201, body PatientOut . Declared status codes: 201 , 422 .

Sample (real call)

Request (clinical_ops):

```
curl -X POST -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" -d '{"patient_nbr": 990000002, "race": "Caucasian", "gender": "Female"}' "http://127.0.0.1:8000/patients"
```

Response 201 :

```
{
  "patient_nbr": 990000002,
  "race": "Caucasian",
  "gender": "Female",
  "created_at": "2026-10-09T20:57:48",
  "updated_at": "2026-10-09T20:57:48",
  "encounter_count": null
}
```

3.5 GET /patients

List and search patients.

Roles allowed: administrator, clinical_ops

Parameters

Name	In	Type	Required	Notes
q	query	integer (exclusiveMinimum 0)	no	exact patient_nbr
has_multiple_encounters	query	boolean	no	true: more than one active encounter
sort_by	query	one of: patient_nbr, created_at	no	
sort_dir	query	one of: asc, desc	no	
page	query	integer (minimum 1, default 1)	no	1-based page number
page_size	query	integer (minimum 1, maximum 100, default 25)	no	rows per page, 1..100

Response format: success 200, body Page_PatientOut_. Declared status codes: 200, 422.

Sample (real call)

Request (administrator):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/patients?q=99000&page_size=2"
```

Response 200:

```
{
  "items": [],
  "total": 0,
  "page": 1,
  "page_size": 2,
  "pages": 0,
  "meta": {
    "filters": {
      "q": 99000
    },
    "sort_by": "patient_nbr",
    "sort_dir": "asc",
    "dates_simulated": null
  }
}
```

3.6 GET /patients/{patient_nbr}

Get one patient.

Roles allowed: administrator, clinical_ops

Parameters

Name	In	Type	Required	Notes
patient_nbr	path	integer	yes	

Response format: success 200, body PatientOut_. Declared status codes: 200, 422.

Sample (real call)

Request (administrator):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/patients/990000002"
```

Response 200:

```
{
  "patient_nbr": 990000002,
  "race": "Caucasian",
  "gender": "Unknown",
  "created_at": "2026-10-09T20:57:48",
  "updated_at": "2026-10-09T20:57:48",
  "encounter_count": null
}
```

3.7 PUT /patients/{patient_nbr}

Update a patient (partial).

Roles allowed: administrator, clinical_ops

Parameters

Name	In	Type	Required	Notes
patient_nbr	path	integer	yes	

Request body (application/json)

Field	Type	Required
race	string (maxLength 30)	no
gender	one of: Male , Female , Unknown	no

Response format: success 200 , body PatientOut . Declared status codes: 200 , 422 .

Sample (real call)

Request (clinical_ops):

```
curl -X PUT -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" -d '{"gender": "Unknown"}' "http://127.0.0.1:8000/patients/990000002"
```

Response 200:

```
{
  "patient_nbr": 990000002,
  "race": "Caucasian",
  "gender": "Unknown",
  "created_at": "2026-10-09T20:57:48",
  "updated_at": "2026-10-09T20:57:48",
  "encounter_count": null
}
```

3.8 DELETE /patients/{patient_nbr}

Soft-delete a patient (administrator only).

Roles allowed: administrator

Parameters

Name	In	Type	Required	Notes
patient_nbr	path	integer	yes	

Response format: success 204, no body. Declared status codes: 204, 422.

Sample (real call)

Request (administrator):

```
curl -X DELETE -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/patients/990000002"
```

Response 204:

(empty body)

3.9 GET /patients/{patient_nbr}/encounters

Active encounters of one patient.

Roles allowed: administrator, clinical_ops

Parameters

Name	In	Type	Required	Notes
patient_nbr	path	integer	yes	
sort_by	query	one of: encounter_id, admission_date, time_in_hospital, num_medications, number_inpatient, age_order	no	
sort_dir	query	one of: asc, desc	no	
page	query	integer (minimum 1, default 1)	no	1-based page number
page_size	query	integer (minimum 1, maximum 100, default 25)	no	rows per page, 1..100

Response format: success 200, body Page_EncounterOut_. Declared status codes: 200, 422.

Sample (real call)

Request (administrator):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/patients/990000002/encounters"
```

Response 200:

```

{
  "items": [
    {
      "encounter_id": 990000002,
      "patient_nbr": 990000002,
      "admission_date": "2005-03-01",
      "discharge_date": "2005-03-04",
      "age_group": "[70-80)",
      "age_order": 8,
      "admission_type_id": 1,
      "admission_type": "Emergency",
      "admission_source_id": 7,
      "admission_source": "Emergency Room",
      "discharge_disposition_id": 1,
      "discharge_disposition": "Discharged to home",
      "medical_specialty": "Cardiology",
      "payer_code": "MC",
      "time_in_hospital": 3,
      "num_lab_procedures": 40,
      "num_procedures": 1,
      "num_medications": 12,
      "number_outpatient": 0,
      "number_emergency": 0,
      "number_inpatient": 1,
      "number_diagnoses": 3,
      "max_glu_serum": "None",
      "a1c_result": "None",
      "med_changed": false,
      "diabetes_med": true,
      "readmitted": "<30",
      "readmitted_30d": true,
      "any_readmission": true,
      "is_readmission_eligible": true,
      "source_batch_id": null,
      "dates_simulated": true
    }
  ],
  "total": 1,
  "page": 1,
  "page_size": 25,
  "pages": 1,
  "meta": {
    "filters": {
      "patient_nbr": 990000002
    },
    "sort_by": "admission_date",
    "sort_dir": "desc",
    "dates_simulated": true
  }
}

```

3.10 POST /encounters

Create an encounter.

Roles allowed: administrator, clinical_ops

Request body (application/json)

Field	Type	Required
encounter_id	integer (maximum 9.223372036854776e+18, exclusiveMinimum 0.0)	yes
patient_nbr	integer (maximum 9.223372036854776e+18, exclusiveMinimum 0.0)	yes
admission_date	string	yes
discharge_date	string	yes
age_group	one of: [0-10), [10-20), [20-30), [30-40), [40-50), [50-60), [60-70), [70-80), [80-90), [90-100)	yes
admission_type_id	integer	yes
discharge_disposition_id	integer	yes
admission_source_id	integer	yes
medical_specialty	string (maxLength 80)	no
payer_code	string (maxLength 10)	no
time_in_hospital	integer (minimum 1.0, maximum 14.0)	yes
num_lab_procedures	integer (minimum 0.0, maximum 250.0)	yes
num_procedures	integer (minimum 0.0, maximum 20.0)	yes
num_medications	integer (minimum 0.0, maximum 150.0)	yes
number_outpatient	integer (minimum 0.0, maximum 100.0)	yes
number_emergency	integer (minimum 0.0, maximum 150.0)	yes
number_inpatient	integer (minimum 0.0, maximum 50.0)	yes
number_diagnoses	integer (minimum 0.0, maximum 30.0)	yes
max_glu_serum	one of: None, Norm, >200, >300	yes
a1c_result	one of: None, Norm, >7, >8	yes
med_changed	boolean	yes
diabetes_med	boolean	yes
readmitted	one of: NO, >30, <30	yes
diagnoses	array of DiagnosisIn	no
medications	array of MedicationIn	no

Response format: success 201, body EncounterOut. Declared status codes: 201, 422.

Sample (real call)

Request (clinical_ops):

```
curl -X POST -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" -d '{"encounter_id": 990000002, "patient_nbr": 990000002, "admission_date": "2005-03-01", "discharge_date": "2005-03-04", "age_group": "[70-80)", "admission_type_id": 1, "discharge_disposition_id": 1, "admission_source_id": 7, "medical_specialty": "Cardiology", "payer_code": "MC", "time_in_hospital": 3, "num_lab_procedures": 40, "num_procedures": 1, "num_medications": 12, "number_outpatient": 0, "number_emergency": 0, "number_inpatient": 0, "number_diagnoses": 3, "max_glu_serum": "None", "a1c_result": "None", "med_changed": false, "diabetes_med": true, "readmitted": "NO", "diagnoses": [{"position": 1, "icd9_code": "250.83"}, {"position": 2, "icd9_code": "401.9"}], "medications": [{"drug_name": "insulin", "dosage_status": "Steady"}]}' 'http://127.0.0.1:8000/encounters'
```

Response 201:

```

{
  "encounter_id": 990000002,
  "patient_nbr": 990000002,
  "admission_date": "2005-03-01",
  "discharge_date": "2005-03-04",
  "age_group": "[70-80)",
  "age_order": 8,
  "admission_type_id": 1,
  "admission_type": "Emergency",
  "admission_source_id": 7,
  "admission_source": "Emergency Room",
  "discharge_disposition_id": 1,
  "discharge_disposition": "Discharged to home",
  "medical_specialty": "Cardiology",
  "payer_code": "MC",
  "time_in_hospital": 3,
  "num_lab_procedures": 40,
  "num_procedures": 1,
  "num_medications": 12,
  "number_outpatient": 0,
  "number_emergency": 0,
  "number_inpatient": 0,
  "number_diagnoses": 3,
  "max_glu_serum": "None",
  "a1c_result": "None",
  "med_changed": false,
  "diabetes_med": true,
  "readmitted": "NO",
  "readmitted_30d": false,
  "any_readmission": false,
  "is_readmission_eligible": true,
  "source_batch_id": null,
  "diagnoses": [
    {
      "position": 1,
      "icd9_code": "250.83"
    },
    {
      "position": 2,
      "icd9_code": "401.9"
    }
  ],
  "medications": [
    {
      "drug_name": "insulin",
      "dosage_status": "Steady"
    }
  ],
  "dates_simulated": true
}

```

3.11 GET /encounters

List and search encounters.

Roles allowed: administrator, analyst, clinical_ops

Parameters

Name	In	Type	Required	Notes
sort_by	query	one of: encounter_id, admission_date, time_in_hospital, num_medications, number_inpatient, age_order	no	
sort_dir	query	one of: asc, desc	no	
date_from	query	string	no	admission_date >= (inclusive; SIMULATED dates)
date_to	query	string	no	admission_date <= (inclusive)
age_group	query	one of: [0-10), [10-20), [20-30), [30-40), [40-50), [50-60), [60-70), [70-80), [80-90), [90-100)	no	
admission_type_id	query	integer (minimum 1)	no	
admission_source_id	query	integer (minimum 1)	no	
discharge_disposition_id	query	integer (minimum 1)	no	
specialty	query	string (maxLength 80)	no	exact medical specialty name
readmitted	query	one of: NO, >30, <30	no	
min_los	query	integer (minimum 1, maximum 14)	no	
max_los	query	integer (minimum 1, maximum 14)	no	
patient_nbr	query	integer (exclusiveMinimum 0)	no	not allowed for the analyst role
q	query	string (maxLength 19)	no	digits only: exact encounter_id (or patient_nbr); otherwise an ICD-9 prefix match
page	query	integer (minimum 1, default 1)	no	1-based page number
page_size	query	integer (minimum 1, maximum 100, default 25)	no	rows per page, 1..100

Response format: success 200, body Page_EncounterOut_. Declared status codes: 200, 422.

Sample (real call)

Request (administrator):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/encounters?age_group=%5B90-100%29&readmitted=%3C30&sort_by=time_in_hospital&sort_dir=desc&page_size=2"
```

Response 200:

```

{
  "items": [
    {
      "encounter_id": 60241242,
      "patient_nbr": 20081232,
      "admission_date": "2000-10-29",
      "discharge_date": "2000-11-12",
      "age_group": "[90-100)",
      "age_order": 10,
      "admission_type_id": 3,
      "admission_type": "Elective",
      "admission_source_id": 4,
      "admission_source": "Transfer from a hospital",
      "discharge_disposition_id": 18,
      "discharge_disposition": "NULL",
      "medical_specialty": null,
      "payer_code": null,
      "time_in_hospital": 14,
      "num_lab_procedures": 40,
      "num_procedures": 1,
      "num_medications": 11,
      "number_outpatient": 0,
      "number_emergency": 0,
      "number_inpatient": 1,
      "number_diagnoses": 9,
      "max_glu_serum": "None",
      "a1c_result": "None",
      "med_changed": false,
      "diabetes_med": false,
      "readmitted": "<30",
      "readmitted_30d": true,
      "any_readmission": true,
      "is_readmission_eligible": true,
      "source_batch_id": "encounters_batch_000_initial",
      "dates_simulated": true
    },
    {
      "encounter_id": 255609894,
      "patient_nbr": 38190690,
      "admission_date": "2007-04-10",
      "discharge_date": "2007-04-23",
      "age_group": "[90-100)",
      "age_order": 10,
      "admission_type_id": 1,
      "admission_type": "Emergency",
      "admission_source_id": 7,
      "admission_source": "Emergency Room",
      "discharge_disposition_id": 1,
      "discharge_disposition": "Discharged to home",
      "medical_specialty": null,
      "payer_code": "MC",
      "time_in_hospital": 13,
      "num_lab_procedures": 66,
      "num_procedures": 0,
      "num_medications": 29,
      "number_outpatient": 0,
      "number_emergency": 0,
      "number_inpatient": 0,
      "number_diagnoses": 9,
      "max_glu_serum": "None",
      "a1c_result": "None",
      "med_changed": true,
    }
  ]
}

```

```

    "diabetes_med": true,
    "readmitted": "<30",
    "readmitted_30d": true,
    "any_readmission": true,
    "is_readmission_eligible": true,
    "source_batch_id": "encounters_batch_001",
    "dates_simulated": true
  }
],
"total": 310,
"page": 1,
"page_size": 2,
"pages": 155,
"meta": {
  "filters": {
    "age_group": "[90-100)",
    "readmitted": "<30"
  },
  "sort_by": "time_in_hospital",
  "sort_dir": "desc",
  "dates_simulated": true
}
}

```

3.12 GET /encounters/{encounter_id}

Get one encounter with diagnoses and medications.

Roles allowed: administrator, analyst, clinical_ops

Parameters

Name	In	Type	Required	Notes
encounter_id	path	integer	yes	

Response format: success 200, body EncounterOut. Declared status codes: 200, 422.

Sample (real call)

Request (clinical_ops):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/encounters/990000002"
```

Response 200:

```

{
  "encounter_id": 990000002,
  "patient_nbr": 990000002,
  "admission_date": "2005-03-01",
  "discharge_date": "2005-03-04",
  "age_group": "[70-80)",
  "age_order": 8,
  "admission_type_id": 1,
  "admission_type": "Emergency",
  "admission_source_id": 7,
  "admission_source": "Emergency Room",
  "discharge_disposition_id": 1,
  "discharge_disposition": "Discharged to home",
  "medical_specialty": "Cardiology",
  "payer_code": "MC",
  "time_in_hospital": 3,
  "num_lab_procedures": 40,
  "num_procedures": 1,
  "num_medications": 12,
  "number_outpatient": 0,
  "number_emergency": 0,
  "number_inpatient": 1,
  "number_diagnoses": 3,
  "max_glu_serum": "None",
  "a1c_result": "None",
  "med_changed": false,
  "diabetes_med": true,
  "readmitted": "<30",
  "readmitted_30d": true,
  "any_readmission": true,
  "is_readmission_eligible": true,
  "source_batch_id": null,
  "diagnoses": [
    {
      "position": 1,
      "icd9_code": "250.83"
    },
    {
      "position": 2,
      "icd9_code": "401.9"
    }
  ],
  "medications": [
    {
      "drug_name": "insulin",
      "dosage_status": "Steady"
    }
  ],
  "dates_simulated": true
}

```

3.13 PUT /encounters/{encounter_id}

Update an encounter (partial).

Roles allowed: administrator, clinical_ops

Parameters

Name	In	Type	Required	Notes
encounter_id	path	integer	yes	

Request body (application/json)

Field	Type	Required
admission_date	string	no
discharge_date	string	no
age_group	one of: [0-10), [10-20), [20-30), [30-40), [40-50), [50-60), [60-70), [70-80), [80-90), [90-100)	no
admission_type_id	integer	no
discharge_disposition_id	integer	no
admission_source_id	integer	no
medical_specialty	string (maxLength 80)	no
payer_code	string (maxLength 10)	no
time_in_hospital	integer (minimum 1.0, maximum 14.0)	no
num_lab_procedures	integer (minimum 0.0, maximum 250.0)	no
num_procedures	integer (minimum 0.0, maximum 20.0)	no
num_medications	integer (minimum 0.0, maximum 150.0)	no
number_outpatient	integer (minimum 0.0, maximum 100.0)	no
number_emergency	integer (minimum 0.0, maximum 150.0)	no
number_inpatient	integer (minimum 0.0, maximum 50.0)	no
number_diagnoses	integer (minimum 0.0, maximum 30.0)	no
max_glu_serum	one of: None, Norm, >200, >300	no
a1c_result	one of: None, Norm, >7, >8	no
med_changed	boolean	no
diabetes_med	boolean	no
readmitted	one of: NO, >30, <30	no
diagnoses	array of DiagnosisIn	no
medications	array of MedicationIn	no

Response format: success 200, body EncounterOut. Declared status codes: 200, 422.

Sample (real call)

Request (clinical_ops):

```
curl -X PUT -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" -d '{"readmitted": "<30", "number_inpatient": 1}' "http://127.0.0.1:8000/encounters/990000002"
```

Response 200:

```

{
  "encounter_id": 990000002,
  "patient_nbr": 990000002,
  "admission_date": "2005-03-01",
  "discharge_date": "2005-03-04",
  "age_group": "[70-80)",
  "age_order": 8,
  "admission_type_id": 1,
  "admission_type": "Emergency",
  "admission_source_id": 7,
  "admission_source": "Emergency Room",
  "discharge_disposition_id": 1,
  "discharge_disposition": "Discharged to home",
  "medical_specialty": "Cardiology",
  "payer_code": "MC",
  "time_in_hospital": 3,
  "num_lab_procedures": 40,
  "num_procedures": 1,
  "num_medications": 12,
  "number_outpatient": 0,
  "number_emergency": 0,
  "number_inpatient": 1,
  "number_diagnoses": 3,
  "max_glu_serum": "None",
  "a1c_result": "None",
  "med_changed": false,
  "diabetes_med": true,
  "readmitted": "<30",
  "readmitted_30d": true,
  "any_readmission": true,
  "is_readmission_eligible": true,
  "source_batch_id": null,
  "diagnoses": [
    {
      "position": 1,
      "icd9_code": "250.83"
    },
    {
      "position": 2,
      "icd9_code": "401.9"
    }
  ],
  "medications": [
    {
      "drug_name": "insulin",
      "dosage_status": "Steady"
    }
  ],
  "dates_simulated": true
}

```

3.14 DELETE /encounters/{encounter_id}

Soft-delete an encounter (administrator only).

Roles allowed: administrator

Parameters

Name	In	Type	Required	Notes
encounter_id	path	integer	yes	

Response format: success 204, no body. Declared status codes: 204, 422.

Sample (real call)

Request (administrator):

```
curl -X DELETE -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/encounters/990000002"
```

Response 204:

(empty body)

3.15 GET /analytics/summary

Headline KPIs.

Roles allowed: administrator, analyst, clinical_ops

Parameters

Name	In	Type	Required	Notes
date_from	query	string	no	admission_date >= (inclusive; SIMULATED)
date_to	query	string	no	admission_date <= (inclusive)
age_group	query	one of: [0-10), [10-20), [20-30), [30-40), [40-50), [50-60), [60-70), [70-80), [80-90), [90-100)	no	
admission_type_id	query	integer (minimum 1)	no	

Response format: success 200, body SummaryResponse. Declared status codes: 200, 422.

Sample (real call)

Request (analyst):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/analytics/summary"
```

Response 200:

```
{
  "data": {
    "total_encounters": 101766,
    "unique_patients": 71518,
    "avg_length_of_stay": 4.395987,
    "avg_num_medications": 16.021844,
    "eligible_encounters": 99343,
    "readmitted_30d": 11314,
    "readmission_rate_30d": 0.113888,
    "any_readmission_rate": 0.471256
  },
  "meta": {
    "filters": {},
    "denominator": "eligible encounters (is_readmission_eligible = 1: expired and hospice discharges excluded); volumes (encounters, average length of stay) use all active encounters",
    "dates_simulated": true,
    "generated_at": "2026-10-09T20:57:50"
  }
}
```

3.16 GET /analytics/admissions-trend

Encounters and readmission rate per month or year.

Roles allowed: administrator, analyst, clinical_ops

Parameters

Name	In	Type	Required	Notes
granularity	query	one of: month, year	no	
date_from	query	string	no	admission_date >= (inclusive; SIMULATED)
date_to	query	string	no	admission_date <= (inclusive)
age_group	query	one of: [0-10), [10-20), [20-30), [30-40), [40-50), [50-60), [60-70), [70-80), [80-90), [90-100)	no	
admission_type_id	query	integer (minimum 1)	no	

Response format: success 200, body GroupResponse. Declared status codes: 200, 422.

Sample (real call)

Request (analyst):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/analytics/admissions-trend?granularity=year"
```

Response 200 (a list of 10 items is shortened to 2):

```
{
  "data": [
    {
      "id": null,
      "label": "1999",
      "encounters": 8604,
      "eligible_encounters": 8417,
      "readmitted_30d": 942,
      "readmission_rate": 0.111916,
      "small_n": false,
      "avg_length_of_stay": 4.711065
    },
    {
      "id": null,
      "label": "2000",
      "encounters": 9000,
      "eligible_encounters": 8858,
      "readmitted_30d": 1030,
      "readmission_rate": 0.116279,
      "small_n": false,
      "avg_length_of_stay": 4.748778
    }
  ],
  "meta": {
    "filters": {
      "granularity": "year"
    },
    "denominator": "eligible encounters (is_readmission_eligible = 1: expired and hospice discharges excluded); volumes (encounters, average length of stay) use all active encounters",
    "dates_simulated": true,
    "generated_at": "2026-10-09T20:57:50"
  }
}
```

3.17 GET /analytics/length-of-stay

Length of stay by group, with histogram.

Roles allowed: administrator, analyst, clinical_ops

Parameters

Name	In	Type	Required	Notes
group_by	query	one of: age_group, admission_type, admission_source, specialty, overall	no	
date_from	query	string	no	admission_date >= (inclusive; SIMULATED)
date_to	query	string	no	admission_date <= (inclusive)
age_group	query	one of: [0-10), [10-20), [20-30), [30-40), [40-50), [50-60), [60-70), [70-80), [80-90), [90-100)	no	
admission_type_id	query	integer (minimum 1)	no	

Response format: success 200, body LengthOfStayResponse . Declared status codes: 200, 422 .

Sample (real call)

Request (analyst):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/analytics/length-of-stay?group_by=age_group"
```

Response 200 (a list of 10 items is shortened to 2; a list of 14 items is shortened to 2):

```
{
  "data": [
    {
      "id": 1,
      "label": "[0-10)",
      "encounters": 161,
      "avg_length_of_stay": 2.546584,
      "min_length_of_stay": 1,
      "max_length_of_stay": 8,
      "histogram": [
        33,
        53
      ],
      "small_n": false
    },
    {
      "id": 2,
      "label": "[10-20)",
      "encounters": 691,
      "avg_length_of_stay": 3.191027,
      "min_length_of_stay": 1,
      "max_length_of_stay": 14,
      "histogram": [
        143,
        225
      ],
      "small_n": false
    }
  ],
  "meta": {
    "filters": {
      "group_by": "age_group"
    },
    "denominator": "eligible encounters (is_readmission_eligible = 1: expired and hospice discharges excluded); volumes (encounters, average length of stay) use all active encounters",
    "dates_simulated": true,
    "generated_at": "2026-10-09T20:57:51"
  }
}
```

3.18 GET /analytics/readmissions

30-day readmission rate by group.

Roles allowed: administrator, analyst, clinical_ops

Parameters

Name	In	Type	Required	Notes
group_by	query	one of: age_group, admission_type, admission_source, discharge_disposition, specialty, month	no	
date_from	query	string	no	admission_date >= (inclusive; SIMULATED)
date_to	query	string	no	admission_date <= (inclusive)
age_group	query	one of: [0-10), [10-20), [20-30), [30-40), [40-50), [50-60), [60-70), [70-80), [80-90), [90-100)	no	
admission_type_id	query	integer (minimum 1)	no	

Response format: success 200, body GroupResponse. Declared status codes: 200, 422.

Sample (real call)

Request (analyst):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/analytics/readmissions?group_by=age_group"
```

Response 200 (a list of 10 items is shortened to 2):

```
{
  "data": [
    {
      "id": 1,
      "label": "[0-10)",
      "encounters": 161,
      "eligible_encounters": 160,
      "readmitted_30d": 3,
      "readmission_rate": 0.01875,
      "small_n": false,
      "avg_length_of_stay": 2.546584
    },
    {
      "id": 2,
      "label": "[10-20)",
      "encounters": 691,
      "eligible_encounters": 690,
      "readmitted_30d": 40,
      "readmission_rate": 0.057971,
      "small_n": false,
      "avg_length_of_stay": 3.191027
    }
  ],
  "meta": {
    "filters": {
      "group_by": "age_group"
    },
    "denominator": "eligible encounters (is_readmission_eligible = 1: expired and hospice discharges excluded); volumes (encounters, average length of stay) use all active encounters",
    "dates_simulated": true,
    "generated_at": "2026-10-09T20:57:51"
  }
}
```

3.19 GET /analytics/medications

Top drugs, insulin status and A1C vs readmission.

Roles allowed: administrator, analyst, clinical_ops

Parameters

Name	In	Type	Required	Notes
date_from	query	string	no	admission_date >= (inclusive; SIMULATED)
date_to	query	string	no	admission_date <= (inclusive)
age_group	query	one of: [0-10), [10-20), [20-30), [30-40), [40-50), [50-60), [60-70), [70-80), [80-90), [90-100)	no	
admission_type_id	query	integer (minimum 1)	no	

Response format: success 200, body MedicationsResponse. Declared status codes: 200, 422.

Sample (real call)

Request (analyst):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/analytics/medications"
```

Response 200 (a list of 10 items is shortened to 2; a list of 4 items is shortened to 2):

```

{
  "data": {
    "top_drugs": [
      {
        "id": null,
        "label": "insulin",
        "encounters": 54383,
        "eligible_encounters": 52964,
        "readmitted_30d": 6578,
        "readmission_rate": 0.124198,
        "small_n": false,
        "avg_length_of_stay": 4.671956
      },
      {
        "id": null,
        "label": "metformin",
        "encounters": 19988,
        "eligible_encounters": 19844,
        "readmitted_30d": 1938,
        "readmission_rate": 0.097662,
        "small_n": false,
        "avg_length_of_stay": 4.356664
      }
    ],
    "insulin_status": [
      {
        "id": null,
        "label": "No",
        "encounters": 47383,
        "eligible_encounters": 46379,
        "readmitted_30d": 4736,
        "readmission_rate": 0.102115,
        "small_n": false,
        "avg_length_of_stay": 4.079248
      },
      {
        "id": null,
        "label": "Steady",
        "encounters": 30849,
        "eligible_encounters": 30069,
        "readmitted_30d": 3420,
        "readmission_rate": 0.113738,
        "small_n": false,
        "avg_length_of_stay": 4.364193
      }
    ],
    "a1c_result": [
      {
        "id": null,
        "label": "None",
        "encounters": 84748,
        "eligible_encounters": 82509,
        "readmitted_30d": 9641,
        "readmission_rate": 0.116848,
        "small_n": false,
        "avg_length_of_stay": 4.308456
      },
      {
        "id": null,
        "label": "Norm",
        "encounters": 4990,
        "eligible_encounters": 4922,

```

```

        "readmitted_30d": 481,
        "readmission_rate": 0.097725,
        "small_n": false,
        "avg_length_of_stay": 4.924048
    }
]
},
"meta": {
    "filters": {},
    "denominator": "eligible encounters (is_readmission_eligible = 1: expired and hospice discharges excluded); volumes (encounters, average length of stay) use all active encounters",
    "dates_simulated": true,
    "generated_at": "2026-10-09T20:57:53"
}
}

```

3.20 GET /analytics/utilization

Prior inpatient visits vs readmission.

Roles allowed: administrator, analyst, clinical_ops

Parameters

Name	In	Type	Required	Notes
date_from	query	string	no	admission_date >= (inclusive; SIMULATED)
date_to	query	string	no	admission_date <= (inclusive)
age_group	query	one of: [0-10), [10-20), [20-30), [30-40), [40-50), [50-60), [60-70), [70-80), [80-90), [90-100)	no	
admission_type_id	query	integer (minimum 1)	no	

Response format: success 200, body GroupResponse. Declared status codes: 200, 422.

Sample (real call)

Request (analyst):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/analytics/utilization"
```

Response 200 (a list of 4 items is shortened to 2):

```
{
  "data": [
    {
      "id": null,
      "label": "0",
      "encounters": 67630,
      "eligible_encounters": 66245,
      "readmitted_30d": 5690,
      "readmission_rate": 0.085893,
      "small_n": false,
      "avg_length_of_stay": 4.209729
    },
    {
      "id": null,
      "label": "1",
      "encounters": 19521,
      "eligible_encounters": 18984,
      "readmitted_30d": 2516,
      "readmission_rate": 0.132533,
      "small_n": false,
      "avg_length_of_stay": 4.669279
    }
  ],
  "meta": {
    "filters": {},
    "denominator": "eligible encounters (is_readmission_eligible = 1: expired and hospice discharges exclude d); volumes (encounters, average length of stay) use all active encounters",
    "dates_simulated": true,
    "generated_at": "2026-10-09T20:57:54"
  }
}
```

3.21 GET /reports/export

Download a report as CSV or Excel.

Roles allowed: aggregated reports: all roles; `report=encounters`: administrator, clinical_ops

Parameters

Name	In	Type	Required	Notes
<code>report</code>	query	one of: <code>admissions_trend</code> , <code>readmission_summary</code> , <code>length_of_stay_summary</code> , <code>encounters</code>	yes	
<code>format</code>	query	one of: <code>csv</code> , <code>xlsx</code>	no	
<code>date_from</code>	query	string	no	<code>admission_date >=</code> (inclusive; SIMULATED)
<code>date_to</code>	query	string	no	<code>admission_date <=</code> (inclusive)
<code>age_group</code>	query	one of: <code>[0-10)</code> , <code>[10-20)</code> , <code>[20-30)</code> , <code>[30-40)</code> , <code>[40-50)</code> , <code>[50-60)</code> , <code>[60-70)</code> , <code>[70-80)</code> , <code>[80-90)</code> , <code>[90-100)</code>	no	
<code>admission_type_id</code>	query	integer (minimum 1)	no	

Response format: success `200`, no body. Declared status codes: `200`, `403`, `413`, `422`.

Sample (real call)

Request (analyst):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/reports/export?report=readmission_summary&format=csv"
```

Response 200 (first lines of the file):

```
breakdown,label,encounters,eligible_encounters,readmitted_30d,readmission_rate,small_n,avg_length_of_stay
age_group,[0-10),161,160,3,0.01875,False,2.546584
age_group,[10-20),691,690,40,0.057971,False,3.191027
age_group,[20-30),1657,1649,236,0.143117,False,3.564876
age_group,[30-40),3775,3764,424,0.112646,False,3.799735
age_group,[40-50),9685,9607,1024,0.106589,False,4.039649
```

3.22 GET /reports/history

Generated reports (administrator, clinical_ops).

Roles allowed: administrator, clinical_ops

Parameters

Name	In	Type	Required	Notes
report_type	query	one of: admissions_trend, readmission_summary, length_of_stay_summary, encounters	no	
date_from	query	string	no	
date_to	query	string	no	whole day, inclusive
sort_by	query	one of: generated_at, report_id	no	
sort_dir	query	one of: asc, desc	no	
page	query	integer (minimum 1, default 1)	no	1-based page number
page_size	query	integer (minimum 1, maximum 100, default 25)	no	rows per page, 1..100

Response format: success 200, body Page_ReportRunOut_. Declared status codes: 200, 422.

Sample (real call)

Request (clinical_ops):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/reports/history?page_size=2"
```

Response 200:

```
{
  "items": [
    {
      "report_id": 58,
      "report_type": "readmission_summary",
      "file_path": "readmission_summary_20261009_205754.csv",
      "row_count": 254,
      "filters_json": {
        "format": "csv"
      },
      "generated_at": "2026-10-09T20:58:01",
      "triggered_by": "docs_analyst_73ac07"
    },
    {
      "report_id": 57,
      "report_type": "readmission_summary",
      "file_path": "readmission_summary_20261009_205724.csv",
      "row_count": 254,
      "filters_json": {
        "format": "csv"
      },
      "generated_at": "2026-10-09T20:57:31",
      "triggered_by": "docs_analyst_e0c3ff"
    }
  ],
  "total": 58,
  "page": 1,
  "page_size": 2,
  "pages": 29,
  "meta": {
    "filters": {},
    "sort_by": "generated_at",
    "sort_dir": "desc",
    "dates_simulated": null
  }
}
```

3.23 GET /admin/audit-logs

Audit trail (administrator).

Roles allowed: administrator

Parameters

Name	In	Type	Required	Notes
user	query	string (maxLength 50)	no	username
action	query	one of: CREATE , UPDATE , DELETE , EXPORT	no	
entity_type	query	string (maxLength 30)	no	
entity_id	query	string (maxLength 40)	no	
date_from	query	string	no	
date_to	query	string	no	whole day, inclusive
sort_by	query	one of: created_at , id	no	
sort_dir	query	one of: asc , desc	no	
page	query	integer (minimum 1, default 1)	no	1-based page number

Name	In	Type	Required	Notes
page_size	query	integer (minimum 1, maximum 100, default 25)	no	rows per page, 1..100

Response format: success 200, body Page_AuditLogOut_. Declared status codes: 200, 422.

Sample (real call)

Request (administrator):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/admin/audit-logs?entity_id=990000002&page_size=5&sort_dir=asc"
```

Response 200 (a list of 4 items is shortened to 2):

```

{
  "items": [
    {
      "id": 57,
      "created_at": "2026-10-09T20:57:48",
      "user_id": 9,
      "username": "docs_clinical_ops_73ac07",
      "role": "clinical_ops",
      "action": "CREATE",
      "entity_type": "patient",
      "entity_id": "990000002",
      "before_json": null,
      "after_json": {
        "race": "Caucasian",
        "gender": "Female",
        "is_deleted": false,
        "patient_nbr": 990000002
      },
      "request_id": "f716ef35a448493c",
      "ip_address": "testclient"
    },
    {
      "id": 58,
      "created_at": "2026-10-09T20:57:48",
      "user_id": 9,
      "username": "docs_clinical_ops_73ac07",
      "role": "clinical_ops",
      "action": "UPDATE",
      "entity_type": "patient",
      "entity_id": "990000002",
      "before_json": {
        "race": "Caucasian",
        "gender": "Female",
        "is_deleted": false,
        "patient_nbr": 990000002
      },
      "after_json": {
        "race": "Caucasian",
        "gender": "Unknown",
        "is_deleted": false,
        "patient_nbr": 990000002
      },
      "request_id": "04dbaea8edb14296",
      "ip_address": "testclient"
    }
  ],
  "total": 4,
  "page": 1,
  "page_size": 5,
  "pages": 1,
  "meta": {
    "filters": {
      "entity_id": "990000002"
    },
    "sort_by": "created_at",
    "sort_dir": "asc",
    "dates_simulated": null
  }
}

```

3.24 GET /admin/pipeline-runs

ETL run history.

Roles allowed: administrator, clinical_ops

Parameters

Name	In	Type	Required	Notes
status	query	one of: RUNNING, SUCCESS, FAILED, SKIPPED_DUPLICATE_FILE	no	
date_from	query	string	no	
date_to	query	string	no	whole day, inclusive
sort_by	query	one of: started_at, run_id	no	
sort_dir	query	one of: asc, desc	no	
page	query	integer (minimum 1, default 1)	no	1-based page number
page_size	query	integer (minimum 1, maximum 100, default 25)	no	rows per page, 1..100

Response format: success 200, body Page_PipelineRunOut_. Declared status codes: 200, 422.

Sample (real call)

Request (clinical_ops):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/admin/pipeline-runs?page_size=2"
```

Response 200:

```
{
  "items": [
    {
      "run_id": 13,
      "source_file": "encounters_batch_003.csv",
      "status": "SUCCESS",
      "started_at": "2026-10-09T20:54:06",
      "finished_at": "2026-10-09T20:54:06",
      "rows_read": 5088,
      "rows_loaded": 23,
      "rows_rejected": 0,
      "rows_skipped_existing": 5065,
      "dq_summary_json": {
        "load": {
          "loaded": 23,
          "outcomes": 23,
          "diagnoses": 67,
          "medications": 22,
          "patients_filled": 0,
          "skipped_existing": 5065,
          "patients_inserted": 16
        },
        "rows_in": 5088,
        "rows_clean": 5088,
        "reject_ratio": 0.0,
        "flagged_count": 6,
        "rows_rejected": 0,
        "corrected_count": 0,
        "counts_per_rule": {
          "DQ01": 0,
          "DQ02": 0,
          "DQ03": 0,
          "DQ04": 0,
          "DQ05": 0,
          "DQ06": 0,
          "DQ07": 0,
          "DQ08": 0,
          "DQ09": 0,
          "DQ10": 0,
          "DQ12": 0,
          "DQ13": 6
        },
        "max_reject_ratio": 0.2,
        "missing_by_column": {
          "age": 0,
          "race": 153,
          "diag_1": 0,
          "diag_2": 6,
          "diag_3": 21,
          "gender": 0,
          "weight": 4932,
          "examide": 0,
          "insulin": 0,
          "acarbose": 0,
          "miglitol": 0,
          "glipizide": 0,
          "glyburide": 0,
          "metformin": 0,
          "a1c_result": 0,
          "payer_code": 683,
          "readmitted": 0,
          "tolazamide": 0,

```

```

        "citoglipton": 0,
        "glimepiride": 0,
        "med_changed": 0,
        "nateglinide": 0,
        "patient_nbr": 0,
        "repaglinide": 0,
        "tolbutamide": 0,
        "diabetes_med": 0,
        "encounter_id": 0,
        "pioglitazone": 0,
        "troglitazone": 0,
        "acetoexamide": 0,
        "max_glu_serum": 0,
        "rosiglitazone": 0,
        "admission_date": 0,
        "chlorpropamide": 0,
        "discharge_date": 0,
        "num_procedures": 0,
        "num_medications": 0,
        "number_diagnoses": 0,
        "number_emergency": 0,
        "number_inpatient": 0,
        "time_in_hospital": 0,
        "admission_type_id": 0,
        "medical_specialty": 3440,
        "number_outpatient": 0,
        "num_lab_procedures": 0,
        "admission_source_id": 0,
        "glipizide_metformin": 0,
        "glyburide_metformin": 0,
        "metformin_pioglitazone": 0,
        "metformin_rosiglitazone": 0,
        "discharge_disposition_id": 0,
        "glimepiride_pioglitazone": 0
    }
},
"error_message": null,
"triggered_by": "scan"
},
{
    "run_id": 12,
    "source_file": "severe_faulty_batch.csv",
    "status": "FAILED",
    "started_at": "2026-10-09T00:54:44",
    "finished_at": "2026-10-09T00:54:44",
    "rows_read": 5088,
    "rows_loaded": 0,
    "rows_rejected": 5088,
    "rows_skipped_existing": 0,
    "dq_summary_json": {
        "rows_in": 5088,
        "rows_clean": 3562,
        "reject_ratio": 0.29992138364779874,
        "flagged_count": 2,
        "rows_rejected": 1526,
        "corrected_count": 0,
        "counts_per_rule": {
            "DQ01": 0,
            "DQ02": 0,
            "DQ03": 0,
            "DQ04": 1526,
            "DQ05": 0,
            "DQ06": 0,

```

```
"DQ07": 0,
"DQ08": 0,
"DQ09": 0,
"DQ10": 0,
"DQ12": 0,
"DQ13": 2
},
"max_reject_ratio": 0.2,
"missing_by_column": {
  "age": 0,
  "race": 156,
  "diag_1": 1,
  "diag_2": 6,
  "diag_3": 23,
  "gender": 0,
  "weight": 4929,
  "examide": 0,
  "insulin": 0,
  "acarbose": 0,
  "miglitol": 0,
  "glipizide": 0,
  "glyburide": 0,
  "metformin": 0,
  "a1c_result": 0,
  "payer_code": 720,
  "readmitted": 0,
  "tolazamide": 0,
  "citoglipton": 0,
  "glimepiride": 0,
  "med_changed": 0,
  "nateglinide": 0,
  "patient_nbr": 0,
  "repaglinide": 0,
  "tolbutamide": 0,
  "diabetes_med": 0,
  "encounter_id": 0,
  "pioglitazone": 0,
  "troglitazone": 0,
  "acetoexamide": 0,
  "max_glu_serum": 0,
  "rosiglitazone": 0,
  "admission_date": 0,
  "chlorpropamide": 0,
  "discharge_date": 0,
  "num_procedures": 0,
  "num_medications": 0,
  "number_diagnoses": 0,
  "number_emergency": 0,
  "number_inpatient": 0,
  "time_in_hospital": 0,
  "admission_type_id": 0,
  "medical_specialty": 3470,
  "number_outpatient": 0,
  "num_lab_procedures": 0,
  "admission_source_id": 0,
  "glipizide_metformin": 0,
  "glyburide_metformin": 0,
  "metformin_pioglitazone": 0,
  "metformin_rosiglitazone": 0,
  "discharge_disposition_id": 0,
  "glimepiride_pioglitazone": 0
},
"rows_failing_rules": 1526
```

```

    },
    "error_message": "Batch rejected: reject ratio 0.300 exceeds the allowed 0.200 (1526 of 5088 rows)",
    "triggered_by": "scan"
  }
],
"total": 13,
"page": 1,
"page_size": 2,
"pages": 7,
"meta": {
  "filters": {},
  "sort_by": "started_at",
  "sort_dir": "desc",
  "dates_simulated": null
}
}

```

3.25 GET /admin/dq-issues

Row-level data-quality findings.

Roles allowed: administrator, clinical_ops

Parameters

Name	In	Type	Required	Notes
run_id	query	integer (minimum 1)	no	
rule_name	query	string (maxLength 20)	no	
severity	query	one of: error, warning, info	no	
sort_by	query	one of: issue_id, rule_name	no	
sort_dir	query	one of: asc, desc	no	
page	query	integer (minimum 1, default 1)	no	1-based page number
page_size	query	integer (minimum 1, maximum 100, default 25)	no	rows per page, 1..100

Response format: success 200, body PageDqIssueOut_. Declared status codes: 200, 422.

Sample (real call)

Request (clinical_ops):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/admin/dq-issues?page_size=2"
```

Response 200:

```
{
  "items": [
    {
      "issue_id": 1,
      "run_id": 1,
      "encounter_id": 98634858,
      "rule_name": "DQ12",
      "severity": "info",
      "column_name": "gender",
      "bad_value": "Unknown/Invalid",
      "action": "flagged"
    },
    {
      "issue_id": 2,
      "run_id": 1,
      "encounter_id": 226864668,
      "rule_name": "DQ12",
      "severity": "info",
      "column_name": "gender",
      "bad_value": "Unknown/Invalid",
      "action": "flagged"
    }
  ],
  "total": 1900,
  "page": 1,
  "page_size": 2,
  "pages": 950,
  "meta": {
    "filters": {},
    "sort_by": "issue_id",
    "sort_dir": "asc",
    "dates_simulated": null
  }
}
```

3.26 GET /admin/users

List users (never includes password hashes).

Roles allowed: administrator

Parameters

Name	In	Type	Required	Notes
sort_by	query	one of: user_id , username	no	
sort_dir	query	one of: asc , desc	no	
page	query	integer (minimum 1, default 1)	no	1-based page number
page_size	query	integer (minimum 1, maximum 100, default 25)	no	rows per page, 1..100

Response format: success 200 , body Page_AdminUserOut_ . Declared status codes: 200 , 422 .

Sample (real call)

Request (administrator):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/admin/users?page_size=3"
```

Response 200 (a list of 3 items is shortened to 2):

```
{
  "items": [
    {
      "user_id": 1,
      "username": "admin",
      "role": "administrator",
      "is_active": true,
      "last_login_at": "2026-10-09T00:57:27",
      "created_at": "2026-10-08T21:31:48"
    },
    {
      "user_id": 2,
      "username": "clinical_ops",
      "role": "clinical_ops",
      "is_active": true,
      "last_login_at": "2026-10-09T00:22:22",
      "created_at": "2026-10-08T21:31:48"
    }
  ],
  "total": 10,
  "page": 1,
  "page_size": 3,
  "pages": 4,
  "meta": {
    "filters": {},
    "sort_by": "user_id",
    "sort_dir": "asc",
    "dates_simulated": null
  }
}
```

3.27 POST /admin/users

Create a user (administrator).

Roles allowed: administrator

Request body (application/json)

Field	Type	Required
username	string	yes
password	string (minLength 12, maxLength 72)	yes
role	one of: administrator, clinical_ops, analyst	yes

Response format: success 201, body AdminUserOut. Declared status codes: 201, 422.

Sample (real call)

Request (administrator):

```
curl -X POST -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" -d '{"username": "docs_sample_87d180", "password": "<at least 12 characters>", "role": "analyst"}' "http://127.0.0.1:8000/admin/users"
```

Response 201:

```
{
  "user_id": 11,
  "username": "docs_sample_87d180",
  "role": "analyst",
  "is_active": true,
  "last_login_at": null,
  "created_at": "2026-10-09T20:58:01"
}
```

3.28 GET /reference/admission-types

Admission Types.

Roles allowed: any signed-in user

Response format: success 200, body array of RefItem. Declared status codes: 200.

Sample (real call)

Request (analyst):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/reference/admission-types"
```

Response 200 (a list of 8 items is shortened to 2):

```
[
  {
    "id": 1,
    "label": "Emergency"
  },
  {
    "id": 2,
    "label": "Urgent"
  }
]
```

3.29 GET /reference/admission-sources

Admission Sources.

Roles allowed: any signed-in user

Response format: success 200, body array of RefItem. Declared status codes: 200.

Sample (real call)

Request (analyst):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/reference/admission-sources"
```

Response 200 (a list of 25 items is shortened to 2):

```
[
  {
    "id": 1,
    "label": "Physician Referral"
  },
  {
    "id": 2,
    "label": "Clinic Referral"
  }
]
```

3.30 GET /reference/discharge-dispositions

Discharge Dispositions.

Roles allowed: any signed-in user

Response format: success 200, body array of `DispositionItem`. Declared status codes: 200.

Sample (real call)

Request (analyst):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/reference/discharge-dispositions"
```

Response 200 (a list of 30 items is shortened to 2):

```
[
  {
    "id": 1,
    "label": "Discharged to home",
    "is_expired_or_hospice": false
  },
  {
    "id": 2,
    "label": "Discharged/transferred to another short term hospital",
    "is_expired_or_hospice": false
  }
]
```

3.31 GET /reference/specialties

Specialties.

Roles allowed: any signed-in user

Response format: success 200, body array of `RefItem`. Declared status codes: 200.

Sample (real call)

Request (analyst):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/reference/specialties"
```

Response 200 (a list of 72 items is shortened to 2):

```
[
  {
    "id": 1,
    "label": "AllergyandImmunology"
  },
  {
    "id": 2,
    "label": "Anesthesiology"
  }
]
```

3.32 GET /reference/age-groups

Age groups in clinical order (id = age_order).

Roles allowed: any signed-in user

Response format: success 200, body array of RefItem. Declared status codes: 200.

Sample (real call)

Request (analyst):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/reference/age-groups"
```

Response 200 (a list of 10 items is shortened to 2):

```
[
  {
    "id": 1,
    "label": "[0-10]"
  },
  {
    "id": 2,
    "label": "[10-20]"
  }
]
```

4. Role and error examples (real calls)

4.1 Same request as an analyst: the response is data-minimised (no patient_nbr, race or gender)

Request (analyst):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/encounters?age_group=%5B90-100%29&readmitted=%3C30&sort_by=time_in_hospital&sort_dir=desc&page_size=2"
```

Response 200 :

```

{
  "items": [
    {
      "encounter_id": 60241242,
      "admission_date": "2000-10-29",
      "discharge_date": "2000-11-12",
      "age_group": "[90-100)",
      "age_order": 10,
      "admission_type_id": 3,
      "admission_type": "Elective",
      "admission_source_id": 4,
      "admission_source": "Transfer from a hospital",
      "discharge_disposition_id": 18,
      "discharge_disposition": "NULL",
      "medical_specialty": null,
      "payer_code": null,
      "time_in_hospital": 14,
      "num_lab_procedures": 40,
      "num_procedures": 1,
      "num_medications": 11,
      "number_outpatient": 0,
      "number_emergency": 0,
      "number_inpatient": 1,
      "number_diagnoses": 9,
      "max_glu_serum": "None",
      "a1c_result": "None",
      "med_changed": false,
      "diabetes_med": false,
      "readmitted": "<30",
      "readmitted_30d": true,
      "any_readmission": true,
      "is_readmission_eligible": true,
      "dates_simulated": true
    },
    {
      "encounter_id": 255609894,
      "admission_date": "2007-04-10",
      "discharge_date": "2007-04-23",
      "age_group": "[90-100)",
      "age_order": 10,
      "admission_type_id": 1,
      "admission_type": "Emergency",
      "admission_source_id": 7,
      "admission_source": "Emergency Room",
      "discharge_disposition_id": 1,
      "discharge_disposition": "Discharged to home",
      "medical_specialty": null,
      "payer_code": "MC",
      "time_in_hospital": 13,
      "num_lab_procedures": 66,
      "num_procedures": 0,
      "num_medications": 29,
      "number_outpatient": 0,
      "number_emergency": 0,
      "number_inpatient": 0,
      "number_diagnoses": 9,
      "max_glu_serum": "None",
      "a1c_result": "None",
      "med_changed": true,
      "diabetes_med": true,
      "readmitted": "<30",
      "readmitted_30d": true,
    }
  ]
}

```

```

    "any_readmission": true,
    "is_readmission_eligible": true,
    "dates_simulated": true
  }
],
"total": 310,
"page": 1,
"page_size": 2,
"pages": 155,
"meta": {
  "filters": {
    "age_group": "[90-100)",
    "readmitted": "<30"
  },
  "sort_by": "time_in_hospital",
  "sort_dir": "desc",
  "dates_simulated": true
}
}

```

4.2 An analyst calling a patient endpoint is refused (403), whatever the interface hides

Request (analyst):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/patients"
```

Response 403:

```

{
  "error": {
    "code": "forbidden",
    "message": "You do not have permission to perform this action",
    "details": null
  }
}

```

4.3 clinical_ops cannot delete (403): only administrators can

Request (clinical_ops):

```
curl -X DELETE -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/encounters/990000002"
```

Response 403:

```

{
  "error": {
    "code": "forbidden",
    "message": "You do not have permission to perform this action",
    "details": null
  }
}

```

4.4 A soft-deleted encounter is invisible: 404, like an id that never existed

Request (administrator):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/encounters/990000002"
```

Response 404:

```
{
  "error": {
    "code": "not_found",
    "message": "Encounter 990000002 not found",
    "details": null
  }
}
```

4.5 No token: 401

Request ((no token)):

```
curl -X GET "http://127.0.0.1:8000/encounters"
```

Response 401:

```
{
  "error": {
    "code": "unauthorized",
    "message": "Not authenticated",
    "details": null
  }
}
```

4.6 Invalid body: 422 with field-level details

Request (clinical_ops):

```
curl -X POST -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" -d '{"encounter_id": 990000003, "patient_nbr": 990000002, "admission_date": "2005-03-01", "discharge_date": "2005-02-01", "age_group": "[70-80)", "admission_type_id": 1, "discharge_disposition_id": 1, "admission_source_id": 7, "medical_specialty": "Cardiology", "payer_code": "MC", "time_in_hospital": 40, "num_lab_procedures": 40, "num_procedures": 1, "num_medications": 12, "number_outpatient": 0, "number_emergency": 0, "number_inpatient": 0, "number_diagnoses": 3, "max_glu_serum": "None", "a1c_result": "None", "med_changed": false, "diabetes_med": true, "readmitted": "NO", "diagnoses": [{"position": 1, "icd9_code": "250.83"}, {"position": 2, "icd9_code": "401.9"}], "medications": [{"drug_name": "insulin", "dosage_status": "Steady"}]}' "http://127.0.0.1:8000/encounters"
```

Response 422:

```
{
  "error": {
    "code": "validation_error",
    "message": "Request validation failed",
    "details": [
      {
        "loc": [
          "body",
          "time_in_hospital"
        ],
        "msg": "Input should be less than or equal to 14",
        "type": "less_than_equal"
      }
    ]
  }
}
```

4.7 Unknown sort column is refused, never passed to SQL: 422

Request (administrator):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/encounters?sort_by=password_hash"
```

Response 422:

```
{
  "error": {
    "code": "validation_error",
    "message": "Request validation failed",
    "details": [
      {
        "loc": [
          "query",
          "sort_by"
        ],
        "msg": "Input should be 'encounter_id', 'admission_date', 'time_in_hospital', 'num_medications', 'number_inpatient' or 'age_order'",
        "type": "literal_error"
      }
    ]
  }
}
```

4.8 An analyst may not export encounter-level data (403)

Request (analyst):

```
curl -X GET -H "Authorization: Bearer $TOKEN" "http://127.0.0.1:8000/reports/export?report=encounters&format=csv"
```

Response 403:

```
{
  "error": {
    "code": "forbidden",
    "message": "You do not have permission to export this report",
    "details": null
  }
}
```